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# How autonomous research shapes scientific understanding

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**Autonomous science promises to translate experimental agency into advances in scientific understanding. Yet how agents organize inquiry and turn experimental experience into knowledge remains poorly understood. Here we introduce ChemWorld, programmable chemical worlds that combine autonomous experimentation with independent control of scientific laws and supplied information. Across chemical systems, model configurations and repeated investigations, we connect research objectives and prior knowledge to experimental choices, evolving judgments and predictions under new interventions. Agents test expectations, revise explanations and establish useful local relationships. These achievements follow distinct research paths, and similar predictive accuracy can rest on different evidence histories. Experimental progress, however, does not consistently translate into reliable generalization: local relationships can receive weight beyond their established scope, while observed regularities may be poorly preserved in subsequent judgments. By tracing these transitions and comparing access to research under matched initial information, we examine how inquiry contributes to scientific understanding. ChemWorld makes autonomous research itself an object of empirical investigation, linking the organization of exploration to the formation, revision and use of scientific knowledge.**

An autonomous researcher does more than act on the world. By choosing experiments and measurements, it also shapes the experience from which its scientific judgments form: which phenomena become visible, which explanations encounter a test and which results guide the next investigation. As AI scientists and self-driving laboratories connect increasingly elaborate

experimental cycles, this organization of inquiry becomes part of the scientific activity delegated to machines[1,2]. The promise of autonomous science therefore concerns both the expansion of experimental capability and the transformation of experimental experience into scientific understanding.

Robot scientists and self-driving laboratories have demonstrated hypothesis testing, materials discovery and reaction optimization[3–6]. Language-model agents extend the connection between scientific instructions, tools and laboratory operations[7,8]. These advances make it possible to investigate a further question: how does an agent's own research change the relationships it uses to explain and predict the world? A successful procedure establishes an experimental achievement, and a correct forecast establishes predictive accuracy. Neither alone reveals the evidence from which a judgment arose or the scope that evidence supports[9,10].

The difficulty is that the evidence available to an autonomous researcher is itself an outcome of its research. An initial explanation can suggest a productive experiment; the observation can support, narrow or displace that explanation; the resulting judgment can then guide another intervention. A sequence of successful experiments may establish a useful local relationship without resolving how it extends beyond the observed conditions. Conversely, an unsuccessful intervention can advance understanding by ruling out an expectation. Studying this interaction requires connecting what the researcher initially had reason to expect, what its experiments actually tested and what it subsequently treated as established[11–13].

We study these connections in ChemWorld. Programmable world laws and independently specified public information allow autonomous experimental histories to be compared while keeping their scientific consequences accessible to evaluation. Cross-system studies establish the breadth of operating and predictive outcomes; a five-model comparison identifies different routes to evidence and extrapolation; repeated investigations with contemporaneous records locate changes in working judgments; and a matched research-opportunity comparison tests the contribution of conducting research. Together, these studies follow inquiry from its organization through the formation of relationships to their use under new interventions. They examine both productive scientific activity and departures from evidence, without requiring these outcomes to share a single mechanism.

## Making autonomous scientific inquiry experimentally tractable

ChemWorld combines materials, operations and instruments into stateful virtual chemical systems governed by thermodynamic, kinetic and coupled process relationships. Reaction, thermal processing, phase equilibrium, separation and crystallization components can be connected into multistage procedures. Sample states persist, so an upstream reaction or thermal history can change a later separation or measurement. Agents choose quantities, processing sequences, measurements, repetitions and subsequent experiments within a fixed resource allowance. Their choices determine both the procedures they develop and the observations from which they reason. An evaluator independently controls world parameters and public information, records committed operations and obtains withheld reference outcomes (Fig. 1). The agent does not receive those outcomes during research or assessment.

This arrangement makes the agent's research process experimentally accessible. Knowledge tests and supplied-data tasks assess important scientific abilities[14,15], and interactive discovery environments already connect action to observation[16–19]. ChemWorld connects an agent's evidence selection to evolving public judgments and their later use in composable chemical processes. An attempted operation can be rejected before changing the sample; an observation can support a relationship locally without establishing it under every compatible intervention. Execution records and independently evaluated world responses allow these distinctions to be checked.

The broad study comprises 240 Sol campaigns across electrochemistry, partitioning, reaction processing, aqueous equilibrium, reaction-coupled crystallization and purification, plus 60 equilibrium campaigns from four other model configurations. The systems support different scientific questions: developing an effective electrical or thermal procedure, characterizing a concentration response, or recovering material while respecting quality constraints. Discovery and optimization objectives, partial prior information and experimental budgets provide complementary comparisons within the specified study blocks. Each block uses five fixed worlds and three information conditions: Opaque retains the operational contract without an additional instance dossier; Aligned supplies correct but incomplete information; MisIndexed supplies a specified

mismatch. These manipulations change starting conditions for inquiry while preserving the corresponding experimental interface.

The current process analysis adds temporal resolution through 45 equilibrium sessions and 45 crystallization sessions conducted under a corrected, consistent recovery definition. Each system covers five fixed worlds, three information conditions and three fresh model sessions per condition. These 90 sessions contain 80 complete research-and-assessment chains and 83 valid forecast stages; incomplete sessions remain in the denominator. Earlier crystallization sessions affected by a recovery-description mismatch are retained separately. The broad study establishes cross-system contrasts, whereas the repeated studies examine how particular judgments arise and vary across research histories; their protocols and denominators are not pooled.

Contemporaneous notes connect publicly stated judgments to the observations available at that time. After research, agents seal a scientific account, predict outcomes under twelve new interventions without laboratory access, and reflect without receiving reference truth. Structured process analysis identifies 92 observable judgment–experiment–revision windows across 65 of the current 90 sessions (Extended Data Fig. 1). The other 25 sessions lack a complete window under this recording definition; this does not establish an absence of scientific learning. A separate contemporaneous comparison assigns 30 fresh sessions to research or no research under a common assessment protocol. Across these layers, we distinguish a relationship's agreement with world outcomes, the evidence available to support it and the way the researcher subsequently uses it.

## The self-organization and adaptation of scientific inquiry

Research starts from goals, available information and a constrained action space. The historical electrochemical and reaction studies ask agents either to develop and test scientific understanding or to optimize a public operating score. These commissions lead to genuine experimental achievements, but operating and predictive gains are not interchangeable (Fig. 2a). Optimization produces the better retested electrochemical procedure in 26 of 30 matched comparisons; among those 26, prediction error is lower in thirteen and higher in thirteen. In reaction processing, discovery produces both the better procedure and more accurate score predictions in sev-

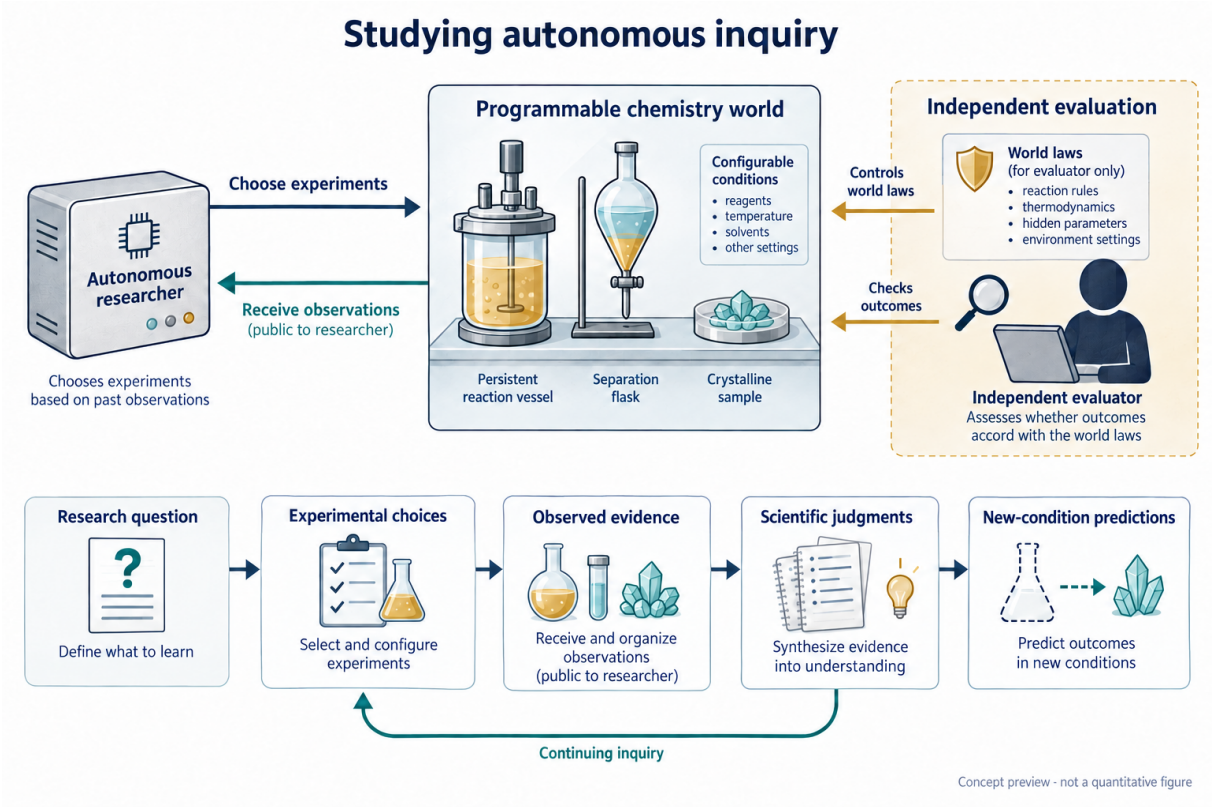



Figure 1 | Making inquiry experimentally tractable. Temporary ImageGen concept illustration. An autonomous researcher selects interventions and receives public observations from a persistent chemical world. The evaluator separately controls hidden parameters and checks world outcomes; reference outcomes are not returned to the researcher. The lower sequence links questions, choices, observations, public judgments and predictions, with judgments informing further choices. It summarizes observable relations, not a direct measurement of private cognition or a demonstrated causal effect of notebook text.

enteen of thirty pairs. The relation between a research objective and its scientific consequences therefore depends on what the inquiry actually establishes.

One electrochemical pair makes the distinction concrete (Fig. 2b). Discovery compares materials and conditions before increasing duration and current cap; optimization screens material combinations and then refines operating conditions for a promising pair. The optimization recommendation retests at 0.756 against discovery's 0.718, but its prediction mean absolute error (MAE) is 0.229 against 0.088. Its later prediction rationale describes several poorly performing cross-pair combinations as low-activity. That account accompanies a forecast of 0.070 against a reference of 0.494 for one combination under new conditions, while another forecast remains accurate at 0.570 against 0.572. The histories expose what a single endpoint obscures: success in developing a procedure can coexist with uneven judgments about the broader experimental space.

Prior information provides another entry into the organization of inquiry. In corrected crystallization, all fifteen aligned-information sessions start with solvent S2 and catalyst C1, whereas all fifteen sessions without an instance dossier start with S0 and C0. Under the specified mismatched dossier, thirteen start with S2 and fourteen with C1. Mean committed solvent coverage is 2.93, 4.00 and 2.40 for Aligned, Opaque and MisIndexed, respectively. These descriptive differences locate how supplied information enters material selection; wider coverage alone does not establish better learning. In reaction processing, partial parameter knowledge has a clear predictive benefit: Aligned reduces macro MAE from 0.102 to 0.057 under discovery and from 0.139 to 0.084 under optimization, improving on Opaque in four of five worlds under each commission (Fig. 2c).

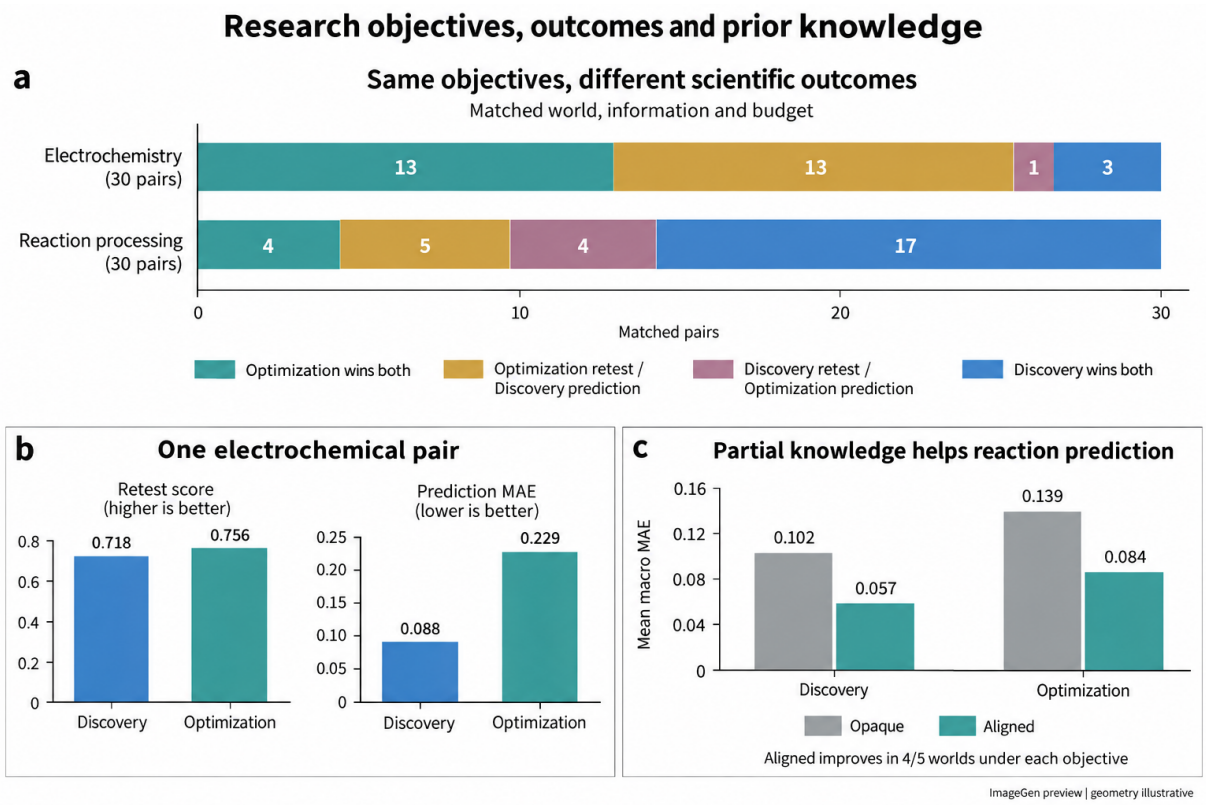



Figure 2 | Research objectives, operating outcomes and prior knowledge. ImageGen data preview of retained cross-system comparisons. a, Thirty matched pairs in each system, classified by which commission has the higher retest score and lower score-prediction MAE. Counts in the four displayed categories are 13/13/1/3 for electrochemistry and 4/5/4/17 for reaction processing. b, One electrochemical pair achieves opposite rankings on retest score and predictive error. c, Five-world mean reaction macro MAE under Opaque and Aligned parameter information, separately for discovery and optimization. Pairing fixes world, information, budget and prior locus within the corresponding block. These are distinct comparisons, not stages of one campaign; no confidence intervals or significance claims are implied. Exact values remain in the historical supplementary tables; generated geometry is illustrative.

Research paths also differ under the same objective and scientific conditions. In the five-model equilibrium comparison, researchers can choose loading, dilution and addition order before predicting pH, dissociation and precipitation. The hidden laws produce a near-constant dissociation fraction over the higher-concentration region, with a changed response at extreme dilution. Eleven of Astra's fifteen campaigns assay positively loaded samples at nominal concentrations no greater than 1 mM, compared with none of fifteen for each of Sol, GPT-5.5, Luna and Terra (Fig. 3a). This difference describes which evidence these particular campaigns acquire, rather than a fixed exploratory trait of a model.

An Astra campaign illustrates the scientific use of newly acquired evidence (Fig. 3c). Its first three assays show dissociation of 6.3–7.4%; batch five tests 18.5 micromolar and observes 53.8%. The later scientific account distinguishes the high-concentration plateau from the dilute response, and the sealed forecast at 13.3 micromolar is 58.80% with an 80% interval of 54.20–63.40%, against a reference mean of 57.97%. Sol/Opaque takes a different route: despite remaining outside the dilute regime during research, its five-world dilute macro MAE is 0.01833 (Fig. 3d). Accurate prediction can therefore accompany either a newly observed change or extrapolation beyond local observations. Direct sampling of every target condition is not a prerequisite for successful inference.

These cross-system and model comparisons establish variation in the evidence histories that precede a judgment. Temporal records then allow us to ask how feedback enters the next research choice. Some campaigns continue a scan planned at the outset; others repeat a borderline recipe, narrow a parameter interval or formulate a new test. Counts of explored materials cannot distinguish these activities. Historical electrochemical discovery and optimization campaigns, for example, cover similar numbers of solvents and both cover all four at the larger budget, despite their differing operating and predictive outcomes.

A single trajectory from the new opportunity comparison, here called equilibrium study A, makes this transition visible (Fig. 4). Under Opaque information, with no instance-specific archival relationship, the initial note predicts a weak-acid trend: increasing concentration should lower pH and dissociation fraction. The first six experiments instead yield dissociation fractions near 0.064–0.079. The next note withdraws support for the expected trend over the measured

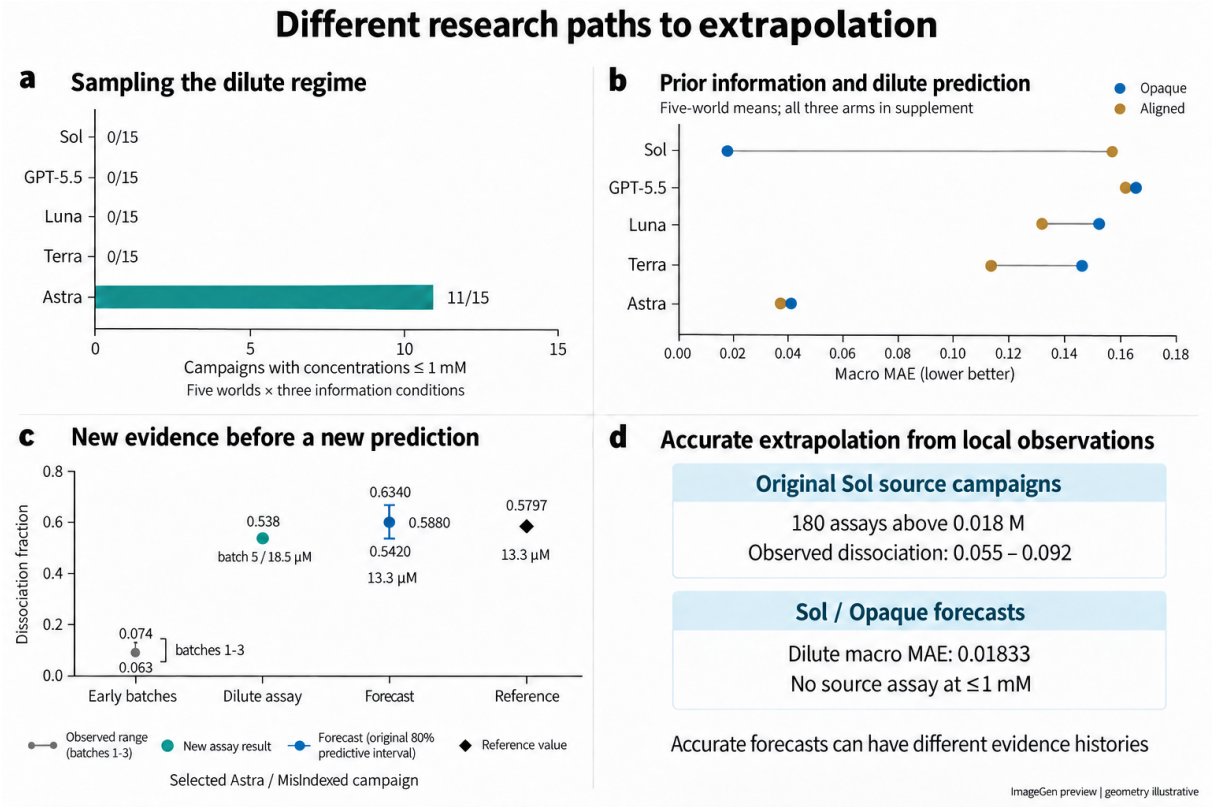



Figure 3 | Different research paths to extrapolation. ImageGen data preview of the historical five-model equilibrium comparison. a, Campaigns sampling at or below 1 mM: Astra 11/15 and each other configuration 0/15. b, Opaque and Aligned five-world mean macro errors on the three dilute queries; all three information arms and exact values are retained in supplementary table F3. c, Selected Astra/MisIndexed campaign: the range in batches 1–3 is an observed range, the dilute assay is a single observation, and the forecast interval is the original nominal 80% interval. d, The 180 assays from all fifteen original Sol campaigns lie above 0.018 M with dissociation 0.055–0.092; the separate Sol/Opaque forecast summary concerns its five campaigns. The cases are retrospectively selected, and there is one campaign per model–world–information cell. Generated geometry is illustrative.

range and proposes that another accessible component might be the relevant input. Batches eight to twelve actually use that component, yet the response remains near a plateau. Its sealed account retains the local relationship and leaves the hidden mechanism unresolved. This is a substantive sequence of expectation, contrary observation, a new candidate explanation and an executed test.

Across the corrected crystallization cohort, 52 of 58 selected immediate plans match the ensuing committed actions and six match partly. This measures a public plan–action connection within observable windows, not a discovery rate or an effect of writing the note. Together with the continuous equilibrium case, these records make the adaptation of inquiry observable: an ex-

perimental history can enter a revised expectation and a subsequent action. What those actions establish requires examining the content and scope of the resulting relationships.

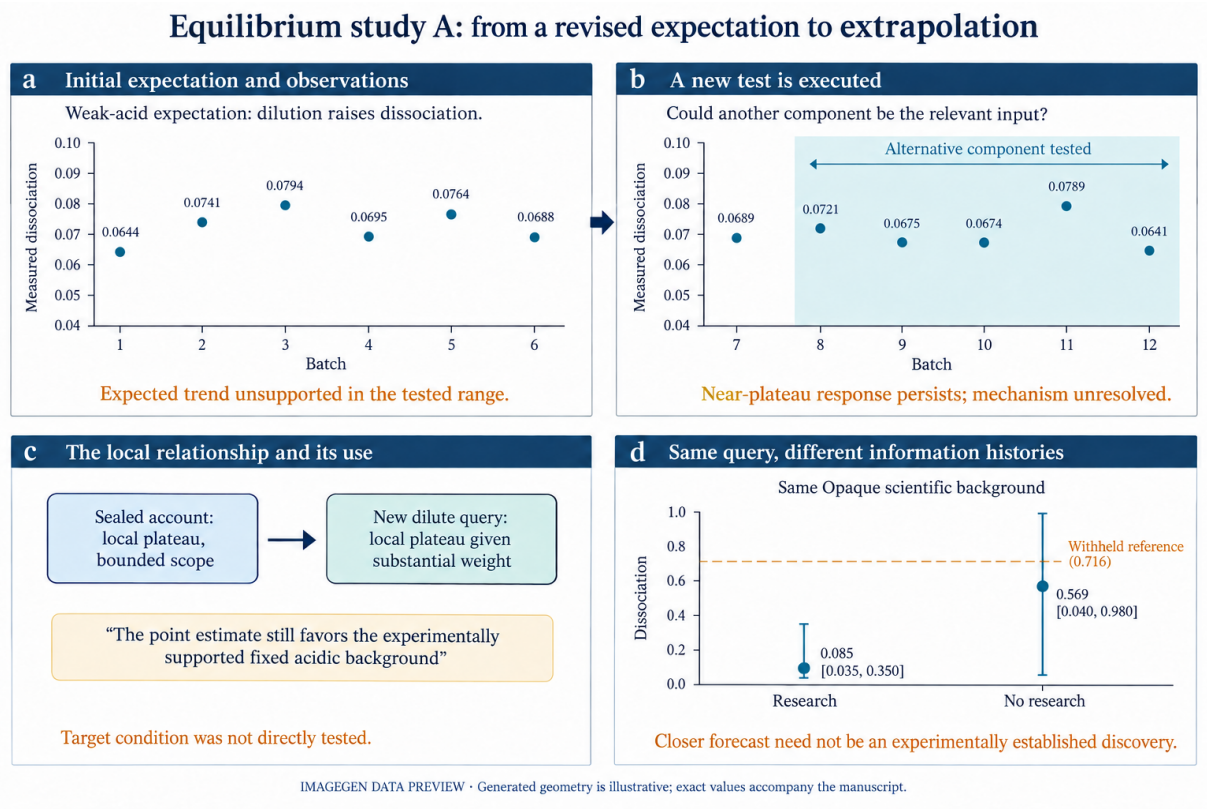



Figure 4 | Equilibrium study A, from a revised expectation to extrapolation. ImageGen data preview of one continuous Research session and its matched NoResearch forecast. a, The first six assays contradict the expected weak-acid trend within the tested range. b, A new candidate explanation motivates alternative-component tests in batches 8–12; the near-plateau persists. c, The bounded relationship in the sealed account is then given substantial weight in a new dilute prediction; the quotation is verbatim. d, Research predicts dissociation 0.085 [0.035, 0.350], NoResearch 0.569 [0.040, 0.980], against withheld mean 0.716. Intervals are agent-stated 80% predictive intervals; source points are individual assays. This is the new opportunity-comparison case, with identity preserved in the evidence archive. Generated geometry is illustrative.

## The formation and evolution of scientific understanding through experimentation

Scientific relationships acquire practical meaning when they support a further test, constrain an explanation or guide a new intervention (Fig. 5). In a corrected crystallization study, a near-threshold fines result leads to a prospective proposal to lower the reagent charge from 0.03 to 0.02 mol. The next batch implements that proposal and reduces fines from 0.529 to 0.387, crossing the 0.50 threshold. The result supports a local recipe improvement. Because the absolute catalyst amount remains fixed, however, the catalyst-to-reagent ratio also changes:

the experiment does not separately identify a concentration mechanism. What is established is a useful relation between the tested recipes and their outcomes, with a narrower scope than a uniquely identified causal law.

A second crystallization study tests an interpolation. After observing different recovery and fines outcomes at nearby solvent volumes, the researcher proposes that 0.065 L should preserve the recovery benefit while improving fines. The next experiment instead increases fines from 0.512 to 0.601, and the final note retracts the expectation. An unsuccessful operating intervention therefore produces an explicit, evidence-responsive revision. Together with the successful lower-charge test, it shows two ways experimentation can advance a working judgment: establishing a local improvement or withdrawing a prediction the observation does not support.

Whether an intervention actually occurred also determines what can be learned from it. In seed case B, a researcher plans approximately 1 g of cumulative seed addition, but only 0.05 g commits and nineteen subsequent requests are rejected. Its later note correctly recognizes that the high-dose test remains unexecuted. Seed case C encounters analogous rejected requests yet describes 1 g as tested and uses the outcome against a seed-based explanation. These contrasting accounts concern the evidential meaning assigned to the same kind of execution limitation. Additional thermal-history and near-threshold repetition cases remain in the supplementary record.

The same equilibrium study A now connects a warranted local revision to its later use (Fig. 4c,d). When forecasting an extremely dilute condition, the agent writes: “The point estimate still favors the experimentally supported fixed acidic background”. It predicts dissociation of 0.085 with an 80% interval of 0.035–0.350; the withheld reference mean is 0.716. Both its scientific account and forecast recognize that this regime was not tested. The transition is therefore more specific than an omitted caveat: a relationship supported locally receives substantial weight in a prediction whose continuation the experiments have not resolved. The matched no-research forecast for the same query is 0.569, with interval 0.040–0.980, and explicitly relies on an untested prior assumption. Its greater numerical accuracy does not establish a better experimental discovery.

That continuation is not inevitable. Two sessions in another fixed equilibrium world both observe plateaux at relatively high loading, yet their dilute macro errors are 0.16287 and 0.01462. For the same dilute query, they predict dissociation of 0.130 and 0.626 against a reference mean

of 0.65463. The more accurate session uses a weak-acid continuation consistent with its plateau observations and explicitly describes it as conjectural. Together with the historical Sol/Opaque and Astra cases, these records distinguish several routes to a judgment: an observed change, a plausible extrapolation and a local relationship carried into an unresolved regime. Numerical correctness and experimental establishment remain separate questions.

The process evidence thus identifies scientifically consequential transitions rather than a single category of error. A failed experiment can narrow what the researcher has reason to expect; a successful intervention can establish a bounded improvement; and a correct recognition of local regularity can coexist with an unreliable application elsewhere. The point at which a judgment is formed and the point at which it is used need not have the same evidential status. Following both points explains why operational failure need not mean an absence of learning, and why a defensible local finding need not guarantee a defensible extrapolation.

## The nature and conditions of scientific progress in autonomous inquiry

The preceding results locate how research is organized and how relationships acquire practical meaning. We next ask when those achievements carry into reliable judgments under new conditions. Prior knowledge, additional resources and the opportunity to conduct experiments can each contribute, but their benefits need not be uniform across the scientific questions a researcher encounters.

The equilibrium prior comparison contrasts with the positive reaction results. In the original Sol campaigns, correct partial information lowers macro MAE on nine queries from 0.01008 to 0.00465, yet increases it on the three most dilute queries from 0.01833 to 0.15862. Both directions occur in all five worlds. A separate repeated cohort shows the same aggregate directions with different magnitudes: 0.00907 to 0.00512 on nine queries and 0.09880 to 0.13205 on the dilute subset. These cohorts are reported separately. The model comparison further limits the interpretation (Fig. 3b): Luna and Terra improve with Aligned information in both regions, GPT-5.5 changes little, and Astra retains low dilute error across all three arms. The effect of partial knowledge is therefore conditional on both the researcher and the condition being judged; a local benefit does not determine its value elsewhere.

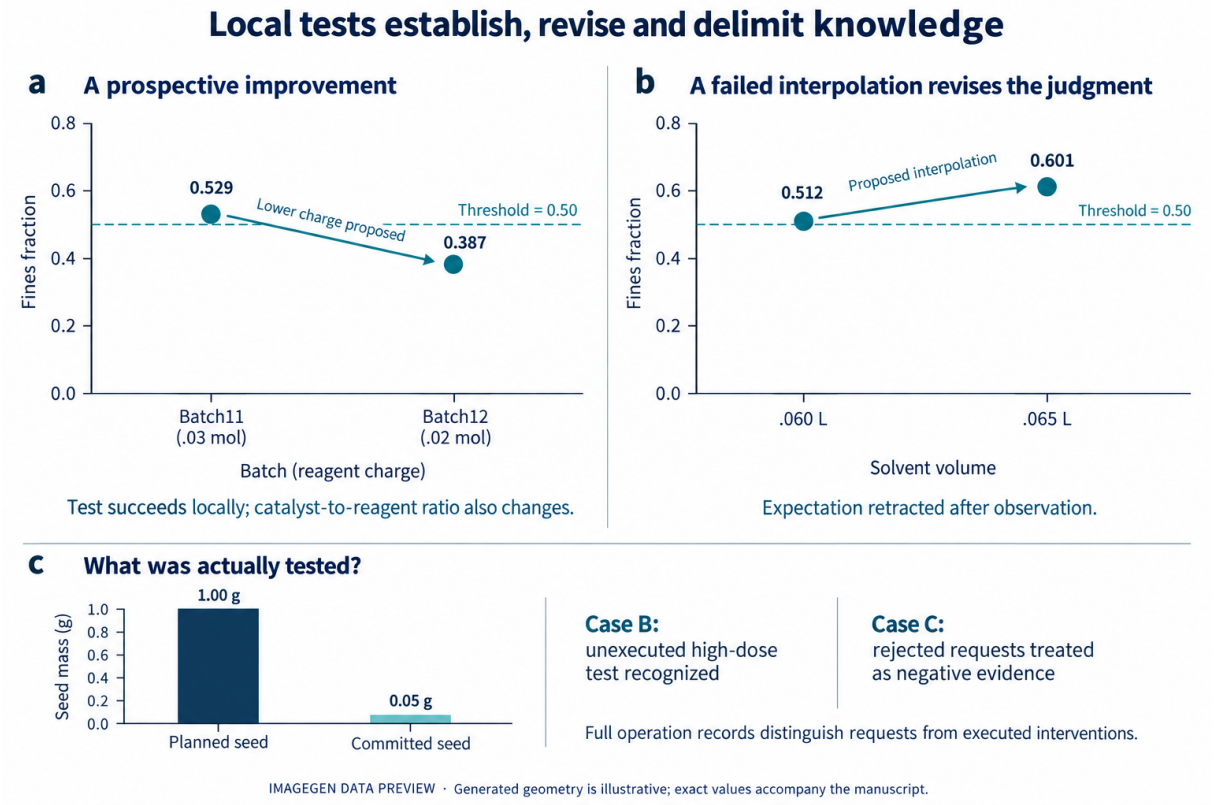



Figure 5 | Local tests establish, revise and delimit knowledge. ImageGen data preview of distinct crystallization studies. a, Lower charge reduces fines from 0.529 to 0.387, crossing the 0.50 threshold while also changing the catalyst-to-reagent ratio. b, A solvent-volume interpolation raises fines from 0.512 to 0.601; the researcher retracts its expectation. Points in a and b are individual assays, not averages with uncertainty intervals. c, Planned 1 g seed additions leave only 0.05 g committed. Case B recognizes an unexecuted high-dose test; case C treats rejected requests as negative evidence. These are separate researchers, not successive states of one trajectory. Generated geometry is illustrative; complete records accompany the manuscript.

Research resources likewise support different kinds of progress. Historical independent twelve- and twenty-four-batch campaigns improve some electrochemical, partitioning and crystallization predictions without a corresponding increase in feasible crystallization recommendations or mean retested recovery. These are comparisons of complete resource envelopes, not matched continuations of a single research history. Combined with the goal contrasts in Fig. 2, they show why neither a more productive procedure nor a larger experimental budget alone specifies what scientific relationships have been learned.

The matched research-opportunity comparison asks a more direct question: does access to research change subsequent prediction under common initial scientific information (Fig. 6)? All thirty sessions use Opaque information: the public task, material identities, measurement scales, legal operations and initial state are available, but no instance-specific archival law or fitted re-

lationship is supplied. Each of five equilibrium worlds receives three fresh sessions with twelve source batches and three without source experiments. Scientific inputs and sealed assessments are matched within each world and repeat. All thirty sessions terminate, all 180 planned source batches complete, and twenty-nine sessions provide valid forecasts. One research session loses its forecast and later stages to a network interruption; its observations and scientific account remain retained.

Research substantially improves prediction overall. Averaging valid sessions within each world and then weighting the five worlds equally, normalized macro mean absolute error falls from 0.132086 without research to 0.035632 with research, a 73.0% reduction. The direction holds in all five worlds. Yet for the three prospectively designated extreme-dilution queries, error rises from 0.061467 to 0.103677, a 68.7% increase, with deterioration in four of five worlds. Over the remaining nine queries it falls from 0.155625 to 0.012950. The comparison concerns the total opportunity to conduct research, including experiment selection, feedback and available recording tools; it does not isolate any one component.

Within this comparison, the same research opportunity improves overall prediction while worsening it in a defined region. All three response-specific point errors increase on the dilute subset, but by different amounts: Research-minus-NoResearch MAE is +0.00777 for normalized pH, +0.10795 for dissociation fraction and +0.01092 for precipitation signal. Dissociation accounts for 85.2% of the macro-error increase. Figure 6 shows individual sessions and world means; the complete session-by-response and query-level results accompany the manuscript. Equilibrium study A supplies one observed route from a valid local finding to a distant judgment. It does not establish that this route explains every session-level difference or that acquiring more evidence must impair extrapolation.

The result extends beyond point error without yielding a universal story of misplaced confidence. At extreme dilution, research produces lower empirical coverage than no research for all three nominal 80% intervals, while the no-research intervals are substantially wider. An interval score penalizing both width and misses favours research for pH, favours no research for dissociation fraction and is similar for precipitation signal. Across all queries, research improves interval scores for all three responses. A conservative sensitivity analysis allowing the missing session’s

macro error to take any value in its legal range does not reverse either the overall benefit or the dilute deterioration (Methods).

Crystallization exposes a complementary pattern in the use of evidence already acquired (Extended Data Fig. 2). For each of 42 valid forecasts, transparent references use only that same session's source observations. Five-world-equal purity error is 0.03853 for the agent and 0.00510 for the overall source mean, which is more accurate in every valid session. Recovery shows the opposite pattern: agent error is 0.10129 against 0.20443, with the agent outperforming the mean in 39 of 42 sessions. Conditioning on the exact committed solvent set preserves both aggregate directions. Size and fines yield 5/42 and 23/42 wins, respectively. These references diagnose how acquired evidence is used across responses; they are neither no-research controls nor alternative autonomous researchers, and do not estimate the causal benefit of crystallization research.

This response-specific comparison complements the equilibrium result. The challenge is not only to anticipate change beyond an observed region, but also to preserve a relationship when it remains useful. Within the same crystallization histories, predictions extract information about varying recovery more effectively than the source mean, yet fail to match that simple reference for stable purity. Successful evidence use in one response does not establish that all acquired relationships constrain subsequent judgments equally. Across objectives, priors, budgets and research access, scientific progress therefore has several observable dimensions: what is achieved experimentally, what the evidence establishes and how well the resulting relationships guide new judgments.

## Discussion

Autonomous scientific inquiry is a process in which agents organize the evidence from which their subsequent judgments form. Across the studies here, research objectives and supplied information enter experimental choices; observations support revised expectations and new tests; and acquired relationships are used with differing success under new interventions. The breadth of the comparisons and the temporal detail of the process records serve complementary purposes. Cross-system results show that experimental achievement and generalization can align

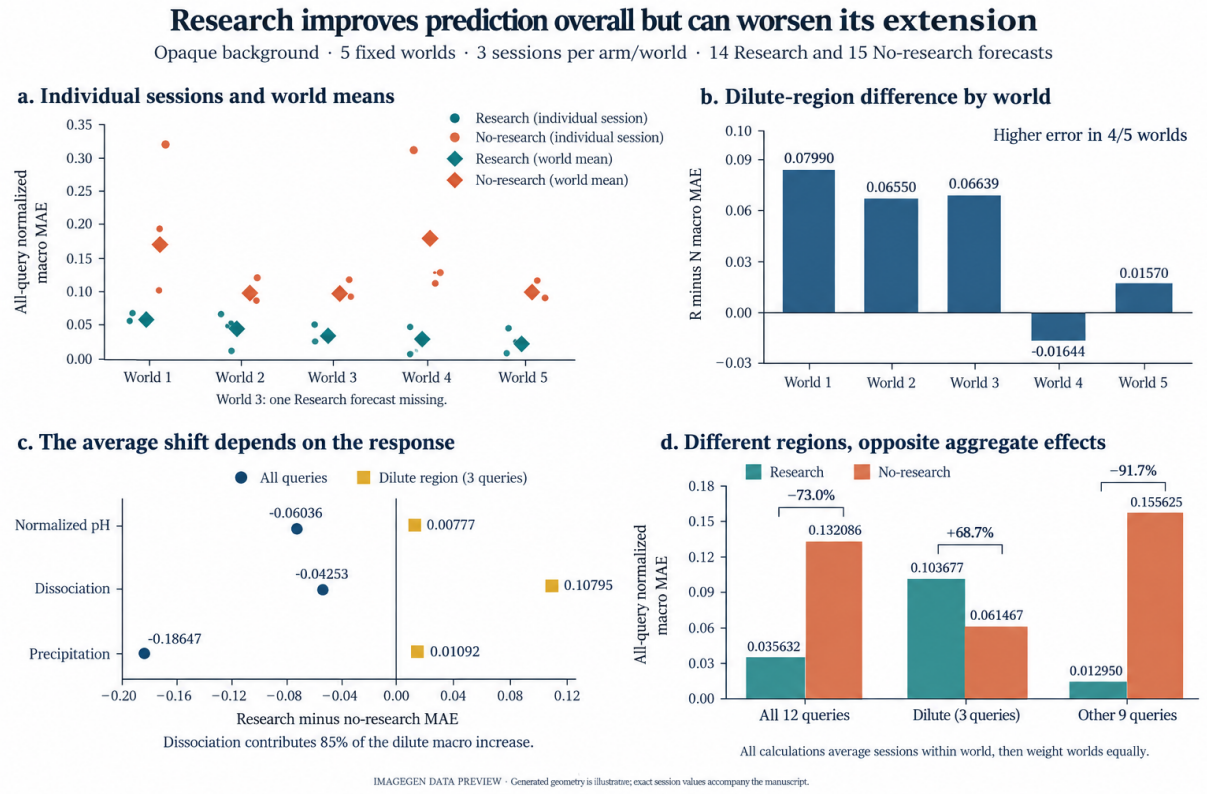



Figure 6 | Overall gains and a regional reversal from research. ImageGen data preview. All thirty sessions use Opaque information and matched scientific inputs. a, Individual all-query macro errors and within-world means; Research has fourteen valid forecasts and NoResearch fifteen. World 3 has one retained missing Research forecast. b, Dilute-region Research-minus-NoResearch macro error is positive in four of five worlds. c, Response-specific shifts differ; dissociation contributes 85.2% of the dilute macro increase. d, Equal-world errors across all twelve, dilute three and other nine queries change by −73.0%, +68.7% and −91.7%, respectively. Generated dots and geometry are illustrative; exact session values, response errors and all 1,044 valid query-response rows are supplied in the evidence archive. No confidence intervals or population sampling inference are implied.

or diverge. Continuous records reveal how such outcomes can coexist with substantive scientific activity, including observing a new response, establishing a local improvement and withdrawing an unsupported expectation.

The findings distinguish truth, evidential support and practical use. A forecast can be accurate even when its explanation remains experimentally unresolved; a locally supported relationship can be applied unsuccessfully elsewhere; and a stable empirical finding can fail to constrain later predictions. The Astra and Sol/Opaque histories make the first distinction visible, equilibrium study A makes the second continuous in time, and the crystallization references expose the third within a common evidence history. These differences would be obscured by assigning each campaign a single label of success or failure. Scientific understanding is better examined

through particular relationships, their support and their uses than inferred solely from an overall operating or prediction score[10,20,21].

These distinctions also explain why discovery and misunderstanding need not be opposing descriptions of an entire researcher. One investigation can establish a useful relationship and still leave its continuation unresolved. A failed test can produce a warranted revision, whereas a successful recipe need not identify the mechanism that made it work. The relevant scientific question is how the status of a relationship changes as evidence arrives and as the relationship is carried into another decision. Experimental design and strong inference provide principles for selecting discriminating tests[11–13]; the present records allow their implementation, limitations and alternatives in autonomous inquiry to be examined rather than presumed.

The controlled comparisons address different parts of this process. The matched opportunity study estimates the contribution of the full research package under a common assessment protocol, finding overall predictive gains together with a regional reversal in the tested worlds. It does not isolate autonomy from feedback, supplied observations or recording tools. A separate fixed-recipe quench intervention identifies physical effects of an operation without providing outcomes to an adapting agent (Extended Data Fig. 3). The process records establish temporal connections between public judgments and actions, but do not independently identify the effect of a particular judgment or a self-confirming sampling loop. Further interventions can distinguish those explanations without making a successful repair strategy a prerequisite for studying how knowledge forms.

The study is limited to selected programmable systems, a small fixed set of worlds and, for repeated process analyses, one model configuration. Fresh sessions are not controlled model-seed replications or independent samples of world laws. Notebook statements are public records rather than access to private cognition; gaps restrict temporal inference. Cases are retrospectively selected from systematically retained records, with single-analyst interpretation and no second-rater reliability estimate. Historical and current protocols retain separate provenance, including the corrected crystallization block and all failures. These limits constrain claims about prevalence and generality while leaving the observed transitions and bounded comparisons available for scrutiny.

ChemWorld brings the organization of inquiry and its epistemic consequences into the same empirical view. It connects a researcher's choices to the evidence obtained, the relationships proposed and the decisions those relationships subsequently inform. This perspective extends evaluation beyond the completion of experimental cycles to the scientific content produced through them. The central question is how autonomous exploration becomes reliable, transferable understanding: which possibilities experiments distinguish, which relationships they support and how those relationships remain responsive to further evidence.

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# Methods

## Worlds, interfaces and experimental units

ChemWorld compiles compatible process components into persistent virtual chemical environments. The public contract specifies materials, legal operations, instruments, resources and task objectives. Private parameters determine each world's responses. Agents access the laboratory and public records, without evaluator state, simulator code, private parameters or withheld reference outcomes. Operations change samples; measurements may consume material. Candidate transitions are validated before commitment. Rejected operations do not constitute the requested physical intervention, although attempts can consume contract-defined allowances. Exact replay checks environment transitions, not regeneration of model decisions.

The broad study covers electrochemistry, partitioning, reaction and thermal processing, aqueous equilibrium, reaction-coupled crystallization and purification. These are controlled virtual systems, not experimentally validated predictions for named physical substances. Current repeated studies use equilibrium and crystallization. Five fixed worlds recur across arms and repetitions; their parameters, initial states and keyed observation mappings are fixed. Repetitions are fresh model sessions, not independent world draws or controlled model-seed replications. The model interface does not expose a usable sampling seed. Repeated identical interventions can share keyed observations, so they are not independent noise samples.

## Evidence layers and retained denominators

The historical broad pool contains 240 Sol-medium campaigns and 60 additional equilibrium campaigns from Luna, Terra, GPT-5.5 and Astra, requested at medium reasoning effort. The five-model equilibrium comparison includes fifteen Sol campaigns already in the 240. Requested model labels do not establish equal realized computation. The original 240 include 60 electrochemical, 30 partitioning, 60 reaction, 45 equilibrium, 30 crystallization and 15 purification campaigns. They retain 3,597/3,600 final assays, all 720 assessment stages and 238 protocol-conforming campaigns under their original rules. The additional sixty retain 720 batches and

240 stages. Two original source-completion failures remain included. Earlier selected notebook studies comprise another 24 sessions and are not pooled into the broad or current process cohort.

The current process cohort combines the 45 equilibrium sessions from the earlier 48-session block and its 42-session extension with a complete replacement block of 45 crystallization sessions under a consistent recovery description. Each system covers five worlds, Opaque/Aligned/MisIndexed information and three sessions per cell, with twelve research batches per session. Equilibrium retains 39 complete chains and 41 valid forecasts; corrected crystallization retains 41 and 42, respectively. The combined current denominator is 90 sessions, 80 complete chains and 83 valid forecasts. Old crystallization sessions remain protocol-affected historical evidence. They are not added to the replacement sessions when calculating current outcomes or process frequencies.

Corrected crystallization has 510/540 source batches, 125/135 assessment stages, 145 notebook versions and 42 recommendation retests. All 45 source trajectories and 42 retests pass replay. Four incomplete chains remain: three lack forecasts, and one retains a valid forecast but lacks the subsequent retrospective. The equilibrium subset and the separate opportunity block retain their own failure tables. Missing outcomes are not set to zero, and later favourable outcomes do not replace completed results. Evidence integration does not change the original development or qualification status of any block.

## Information, autonomy and recording

Historical goal comparisons pair independent discovery and optimization campaigns within world, information arm, batch budget and prior locus. Electrochemistry contributes thirty pairs across twelve- and twenty-four-batch conditions; reaction processing contributes thirty across parameter- and structural-prior blocks at twelve batches. Operating outcomes are independent retests of sealed recommendations. The joint-outcome counts use public-score prediction MAE, whereas the reaction prior comparison uses the mean across its six normalized response errors. These outcomes are not interchanged. Budget comparisons likewise concern independent sessions with complete resource envelopes; they do not estimate the effect of continuing the same investigation.

The historical five-model equilibrium comparison contains five worlds, three information arms and one twelve-batch campaign per cell: seventy-five campaigns, including the fifteen Sol campaigns already counted in the broad study. The 1 mM threshold descriptively summarizes the lowest positively loaded nominal concentration assayed; it is not a discovery criterion. Concentration is based on nominal recipe loading, and the three dilute assessment queries are shared across configurations. Forecast errors, original prediction intervals and acquired evidence are retained for all arms. The selected Astra and electrochemical trajectories are retrospective process illustrations, with scientific accounts written after experimentation; they are not contemporaneous notebook records or additional replications. Complete historical design, response tables and case values remain in Supplementary Information.

Opaque supplies the public task and operational context without the additional instance dossier. Aligned supplies partial information appropriate to the system. In equilibrium, this includes an effective-acidity interval and an anchored local dilution relationship. MisIndexed uses another world's archival relationship with the same format and qualification. In crystallization, Aligned gives nominal descriptors for anonymous materials; MisIndexed swaps solvent S1/S3 descriptors while preserving S0, S2 and catalyst descriptors. Neither dossier specifies the complete world law.

The original equilibrium Opaque/Aligned sessions precede its MisIndexed extension. Corrected crystallization is a later complete block. These descriptive comparisons are not all concurrently randomized treatments, and new versus old crystallization is not an estimate of the causal effect of correcting the prompt. The new opportunity comparison uses a common contemporaneous protocol, described below.

Agents choose operations, quantities, measurements, repetitions and grouping within fixed budgets. Equilibrium permits twelve source batches, 180 operation attempts, twelve intermediate instruments and twelve final assays; crystallization uses its fixed twelve-batch materials envelope and 720-attempt ceiling. A notebook role asks for an initial testable expectation when available; subsequent updates are agent-selected, without a write-every-batch requirement or automatic reinjection. The role may influence research and is not assumed passive. Notes retain version and public-event cursor. These public records do not expose private cognition. Ab-

sence of a notebook-read tool call does not establish absence of access to the conversation's earlier content.

## Assessments and recovery definition

After source research and any operating recommendation, agents seal a scientific account, answer twelve fixed new-intervention queries without laboratory tools and then reflect without truth feedback. Equilibrium includes a structured supplemental assessment. The evidence notation K1/Q/K2/EQS denotes these stages. Queries are hidden during research; post-assessment proposals are not treated as executed source experiments.

Equilibrium errors compare predictions with the mean of five reference observations per query. Coverage checks all five observations. Crystallization uses the noiseless full-process response immediately before final-assay sampling; source observations are noisy. The corrected block retains sixty reference batches, numerically matching the earlier reference outcomes, and separately retests each available committed recommendation.

The corrected recovery definition is seed-excluded recovered solid target divided by the tracked initial reagent charge, in moles, clipped to [0,1]. Source instructions, prediction prompts, observations and scores now refer to this same full-process quantity. The internal crystallization-stage metric instead divides by target product before separation. Earlier process instructions described this latter denominator while observations and scoring used initial reagent charge. Original scores are preserved as protocol-affected diagnostics; they do not enter the corrected cohort. The new block was rerun in full rather than retrospectively relabelling those scores.

Quality requires particles, purity at least 0.80 and fines at most 0.50; joint recommendation success also requires corrected recovery at least 0.10. Corrected joint counts are 13/15 Opaque, 10/14 Aligned and 10/13 MisIndexed among available retests, against fifteen planned sessions in each arm. Particle size is the public index min(d50/250 micrometres, 1); its error is not measured in micrometres.

## Research paths and process interpretation

Retrospective path extraction covers sixty historical electrochemical, forty-five equilibrium and forty-five corrected crystallization sessions. Fifty within-world condition contrasts retain goal, budget, information and cohort. Counts use committed operations; solvent coverage includes additions during resuspension. Rejected attempts do not establish successful material use. Interrupted sessions remain in the planned denominator.

Structured extraction considers all 178 adjacent pairs among 268 notebook versions in the current 90 sessions. Eligibility requires an earlier note, at least one new committed final assay before the next note, and a committed research operation after that later note; measurement and termination alone do not satisfy continuation. All 178 pairs contain new assays. Ninety-two have the required continuation: 34 equilibrium and 58 crystallization windows in 30 and 35 sessions, respectively. The other 86 pairs (44 equilibrium, 42 crystallization) have no later committed research action and remain available in the evidence archive. The 25 sessions without a complete window are not classified as having learned nothing.

Eligibility is computed from event timing before semantic plan matching, without testing whether a proposed action succeeds. Nevertheless, the rule conditions the sample on continued committed research and excludes terminal revisions from this particular action-linked denominator. All 92 eligible windows are coded. Labels concern one selected immediate plan, not every proposal in a note: 83 matched, seven partly matched and two without an explicit next plan; corrected crystallization contributes 52 matched and six partly matched. Full notes retain abandoned, rejected and altered proposals even when another selected action matches. All 182 update/plan quotations and their assay/action cursors are verified against source records. The archive includes every adjacent pair, selected code, full note and execution history.

The same analyst performs coding and internal scientific review. There is no independent second-rater agreement estimate, blinding or claim of adjudicated mechanism prevalence. The team has assigned review to this workflow; completed internal review is distinguished from independent replication. Counterexamples include warranted retractions, correct recognition of

unexecuted tests and accurate but explicitly conjectural extrapolation. Selection of illustrative cases is retrospective.

## Source-only forecast references

Two transparent references are computed for each valid corrected-crystallization forecast using only that researcher's successful source final assays. The first averages all available assays. The second averages assays whose exact set of committed solvent identifiers matches the query, falling back to the overall mean when none match. Solvent sets include successful solvent additions and resuspension, excluding rejected actions. The fitting function receives no test truth. These references are descriptive calculations, not separately trained agents or mechanistically complete baselines.

There are 42 valid sessions and 504 queries. Same-solvent fallback occurs in 186/504 queries: 1/180 Opaque, 95/168 Aligned and 90/156 MisIndexed. Matching does not control thermal history, quantities or final liquid composition. References have no predictive intervals. Each response is reported in its own public scale; no combined score is formed across recovery, purity, size and fines.

## Matched research-opportunity comparison

The fixed design contains five equilibrium worlds, Research/NoResearch and three fresh Sol-medium sessions per cell, for thirty planned sessions. All use Opaque information. The shared public background contains the task and measurement definitions, material catalogue, legal operations/instruments and initial state; the instance archival prior is null. Opaque therefore means absence of an additional instance-specific law, not absence of public chemical framing or model pretraining. The complete initial public contexts accompany the manuscript. Fifteen within-world/repeat checks confirm scientific-input equality after removing only the non-scientific campaign identifier. Research additionally receives twelve autonomous source batches and the existing notebook opportunity; NoResearch receives no source operations or observations. Both arms use identical assessment prompts, schemas, order and restrictions.

All thirty sessions reach terminal status. Research completes 180/180 batches; the combined arms complete 117/120 assessment stages and provide 29 valid forecasts. In world three's third Research repetition, the forecast stream disconnects after the scientific account. Q and the two later stages are missing, without replacement. Thus Research has fourteen valid forecasts, NoResearch fifteen; Research world three has two rather than three. Fifteen source trajectories and twenty-five reference trajectories (300 reference batches) pass exact replay. The retained resource ledger is complete for 29/30 sessions; one has only recorded partial usage. Recorded input plus output totals 13,428,656 tokens. Missing usage is not zero and monetary cost is unavailable.

A first execution attempt failed because the assessment transport configuration omitted a required field. Four launched sessions, two completed source histories totalling twenty-four batches and their resource costs are retained separately; no scientific-account stage completed. After fixing that execution defect, the whole fixed thirty-session block restarted. This was a protocol repair, not replacement based on scientific outcomes. Only the complete restarted block enters the comparison.

The comparison estimates the total research-opportunity package under a shared assessment. It does not isolate autonomy relative to supplied data, feedback alone, notebook use or private representational change. It is between fresh sessions, without a within-session pretest. World and arm order are balanced in a predetermined launch schedule, not a random sample of world laws.

## Quantitative summaries and missingness

Equilibrium macro MAE equally weights normalized pH, dissociation fraction and precipitation signal across the designated queries. The extreme-dilution group comprises three queries fixed before the repeated blocks; the remaining nine are not assumed entirely in distribution. Session errors are averaged within each world, then the five worlds receive equal weight. Individual sessions and all missing stages remain visible. Responses, queries, reference observations and batches are correlated within sessions and worlds; they are not independent researcher samples.

Coverage and mean interval width accompany the proper interval score for the agents' nominal 80% intervals[20]. With lower endpoint l, upper endpoint u, outcome y and alpha = 0.2, the score is (u-l) plus (2/alpha)(l-y) when y is below l, or (2/alpha)(y-u) when y is above u. Lower is better. Aggregates weight worlds equally. Agent intervals are not confidence intervals for our group means.

A post hoc sensitivity analysis assigns the single missing opportunity forecast any macro MAE in [0,1], the legal error range, and recomputes means using the planned three sessions per arm/world and five equal worlds. Research-minus-NoResearch bounds are [-0.098659, -0.031992] overall and [+0.036768, +0.103434] for dilution. These are conservative bounds on missing-outcome influence, not imputation, confidence intervals or population inference. Three repetitions per fixed condition describe variation without establishing a population reliability rate.

## Matched quench intervention

A separate retained block applies the eighteen original first-cohort crystallization recommendations in five worlds, with and without a single quench operation: ninety pairs, 180 trajectories. All other action arguments and initial states are matched. Quench is removed at its recorded location or inserted immediately before the first seed/crystallization-cooling operation. No agent adapts or receives these outcomes.

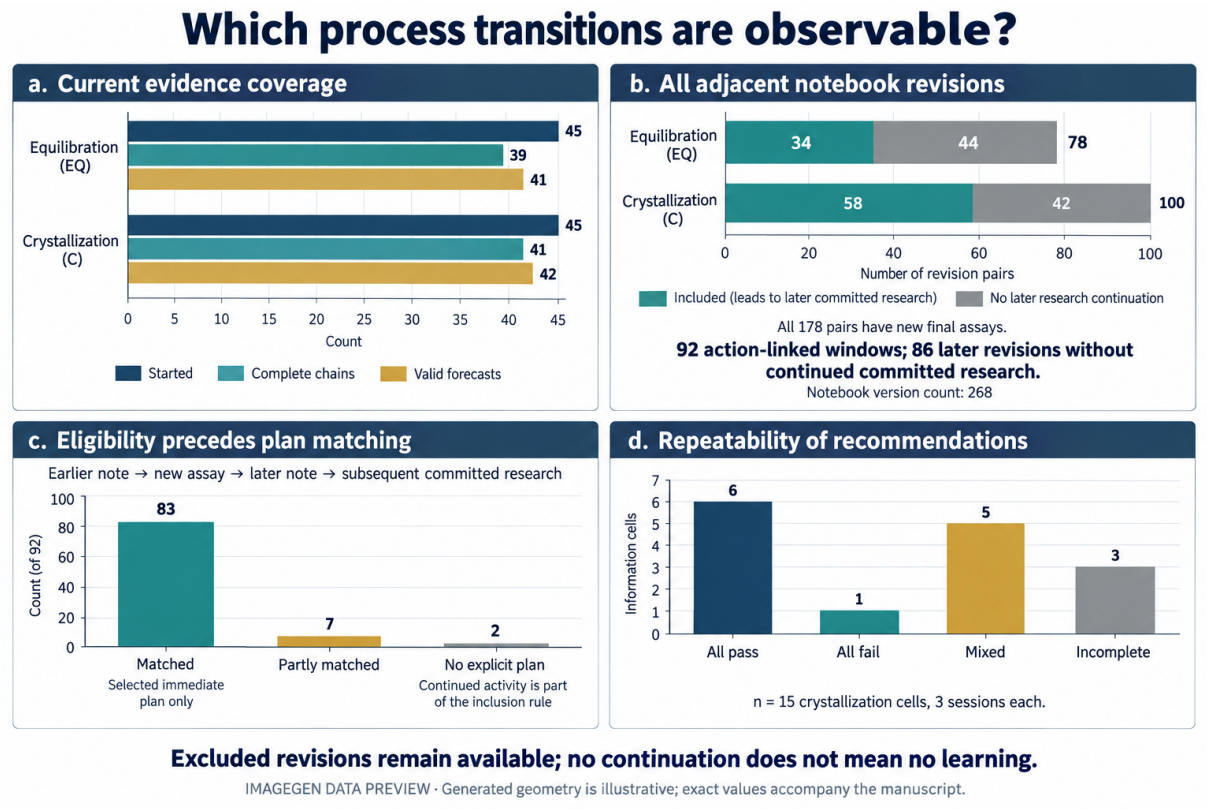
All trajectories are retained and replayed; seventy-five pairs reach both final assays. In five, the absent arm lacks particles; in ten, the present arm makes the subsequent fixed cooling request invalid. Failed arms are not repaired. Among complete pairs, mean present-minus-absent fines changes by +0.30937 and recovery by +0.00741; joint success changes from 34 to 4. These conditional effects describe physical operations under the implemented initial-charge recovery metric. Quench changes multiple physical state variables and does not isolate cooling rate or a cause of scientific judgment. Donor recipes are historical and are not sampled from the corrected block.

## Data and code availability

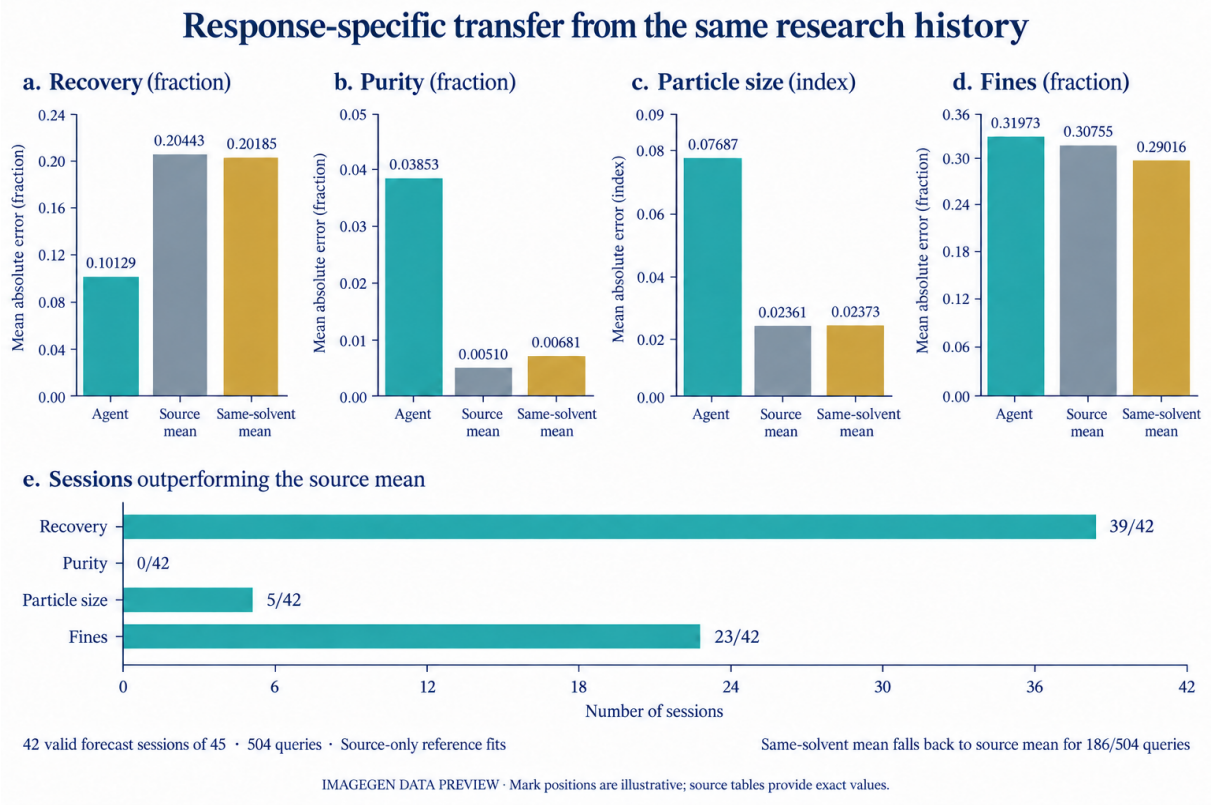
A portable review archive accompanies the manuscript with complete public initial inputs, 1,044 query-response prediction rows, all 270 planned session/group/response cells, world summaries, the continuous focal study, all 178 adjacent note pairs and full public histories for the current 90 process sessions. Missing forecasts remain explicit. Original summaries, exact replay and resource records remain in the project repository; the archive adds no participant evidence or independent second-rater assessment. Supplementary Information reports the numerical decomposition and coding-selection rules. A persistent public deposit remains to be finalized. Manuscript compilation reads retained results without new model or simulator experiments.

Figures are temporary ImageGen previews: one framework illustration and eight data-layout previews, comprising six main figures and three Extended Data figures. Numerical anchors come from retained evidence; generated marks remain illustrative. Exact results are reported in the text and tables. Prompts and provenance accompany the artwork.

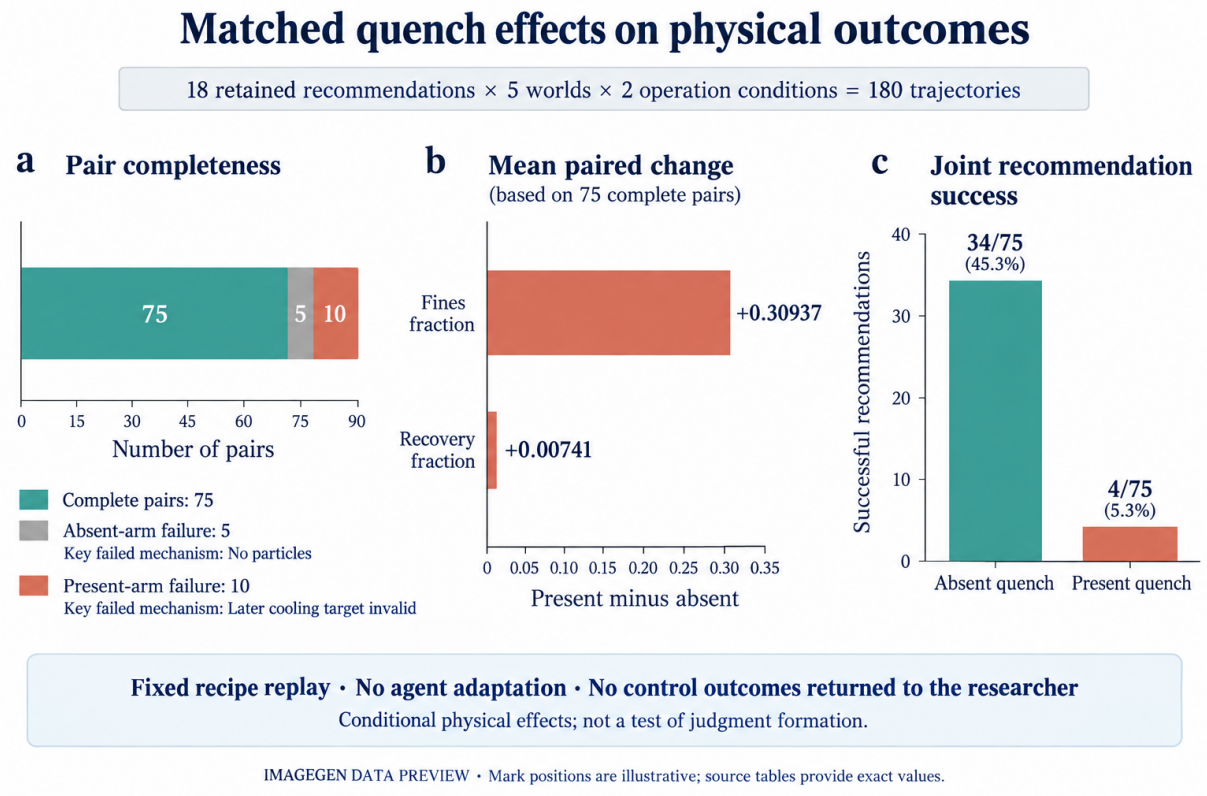
# Extended Data




Extended Data Figure 1 | Which process transitions are observable? ImageGen data preview. a, Current session coverage and valid forecasts. b, All 178 adjacent version pairs contain new assays: 34/78 equilibrium and 58/100 crystallization pairs also have a later committed research action, yielding 92 eligible windows. The remaining 86 revisions are retained without an action-linked code. c, The selected immediate plan is matched in 83 windows, partly matched in seven and absent in two. Eligibility precedes semantic matching but conditions the sample on continued activity; these counts do not measure learning success or cover every proposal. d, Across fifteen crystallization world-by-information cells with three sessions each, six are all-pass, one all-fail, five mixed and three incomplete. Generated geometry is illustrative.




Extended Data Figure 2 | Response-specific transfer from the same research history. ImageGen data preview. a–d, Agent forecasts and two source-only references are compared for recovery, purity, particle-size index and fines, retaining each response's own public scale. Particle size is the dimensionless index min(d50/250 micrometres, 1). e, Agent forecasts outperform the overall source mean in 39/42, 0/42, 5/42 and 23/42 sessions, respectively. The 42 valid corrected-crystallization sessions supply 504 forecast queries. Same-solvent means fall back to the overall mean in 186/504 queries and do not match thermal history or quantities. Neither reference is fitted to held-out truth. Generated geometry is illustrative; five-world-equal numerical errors are reported in the supplementary tables.




Extended Data Figure 3 | Matched quench effects on physical outcomes. ImageGen data preview. Eighteen historical recommendations are replayed in five worlds with and without one quench operation, yielding ninety pairs and 180 trajectories. a, Seventy-five pairs yield both final assays; five absent arms lack particles and ten present arms invalidate a subsequent fixed cooling target. b, Complete-pair mean changes are +0.30937 in fines and +0.00741 in recovery. c, Joint success changes from 34/75 without quench to 4/75 with quench. These are conditional physical effects under the initial-charge recovery definition. The agent neither adapts nor receives control outcomes, so this intervention does not identify judgment formation. Generated geometry is illustrative and all failed pairs remain reported.